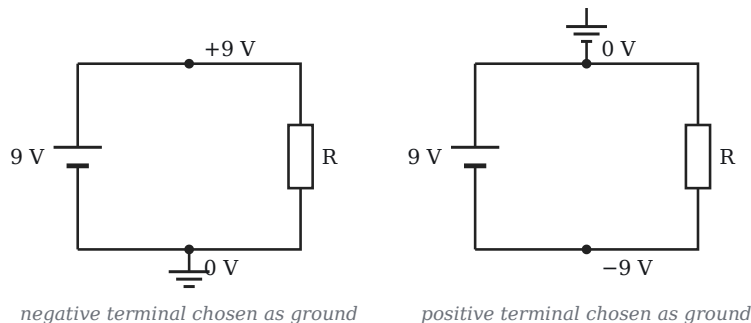


Recitation 2 — Circuits with more than one component, and the start of Boolean algebra

1 Ground and the 0 V reference

Voltage is a difference. Strictly, “the voltage at node A” is meaningless until you say *relative to what*. Circuit ground is that choice: one node you declare to be 0 V, after which every other node has a number.

The choice is free and it changes nothing physical. Currents, and the voltage *across* every component, come out the same whichever node you pick. Only the labels move.

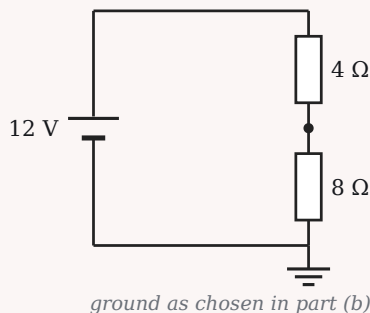


The symbol does two jobs. It marks the reference node, and it lets you stop drawing wires: every point carrying the same ground symbol is understood to be connected to that one node, even with no line between them. That is why real schematics stay readable.

Circuit ground is not Earth ground. Earth ground is a physical connection to the building's protective grounding system, there for safety. A ground symbol in a schematic does not by itself mean the circuit is tied to the Earth.

Problem 1

A 12 V source is in series with a 4Ω resistor (top) and an 8Ω resistor (bottom).

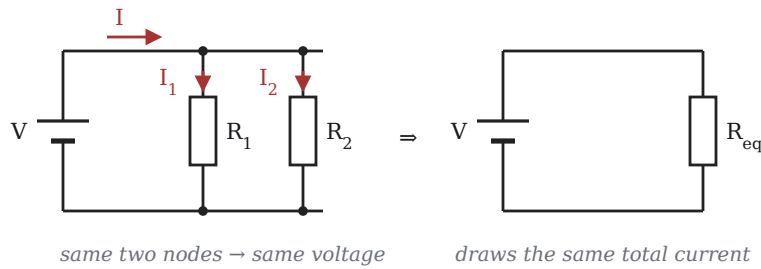


- Find the current and the voltage across each resistor.
- Take the bottom node as 0 V. What is the potential at the middle node and at the top node?
- Now take the *middle* node as 0 V instead. What are the potentials at the top and bottom nodes?
- Which numbers changed, and which physical quantities did not?

2 Resistors in parallel

Parallel resistors sit across the same two nodes, so they see the same V . Each takes the current Ohm's law says it should, and those currents add:

$$I = V/R_1 + V/R_2 = V(1/R_1 + 1/R_2) \Rightarrow 1/R_{eq} = 1/R_1 + 1/R_2 + \dots + 1/R_n$$



For exactly two resistors this rearranges into the form you will use most:

$$R_{eq} = R_1 R_2 / (R_1 + R_2)$$

Three consequences worth memorising, because they let you sanity-check an answer in a second:

- n equal resistors in parallel give R/n .
- The equivalent is always **smaller than the smallest** resistor in the group. If your answer is not, you made an arithmetic error.
- A small resistor in parallel with a much larger one is approximately the small one alone. The large one barely participates.

Careful: the reciprocal is the thing that adds, not the resistance. Writing $1/R_{eq} = 1/R_1 + 1/R_2$ and then forgetting to invert at the end is the single most common slip on this topic.

Problem 2

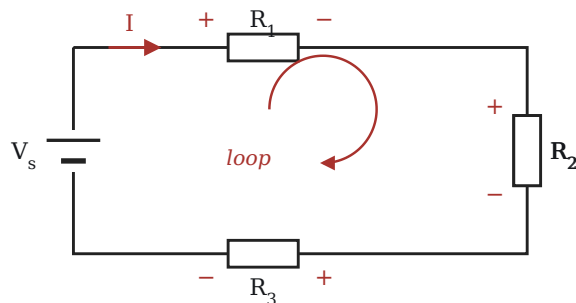
Find the equivalent resistance.

- $6\ \Omega$ in parallel with $3\ \Omega$
- three $3\ \Omega$ resistors in parallel
- $100\ \Omega$ in parallel with $1\ \Omega$ — and comment on the result
- You have a $12\ \Omega$ resistor. What value in parallel with it gives $4\ \Omega$?

3 Series, KVL, and the voltage divider

Kirchhoff's voltage law. Around any closed loop, the algebraic sum of the voltages is zero. It is conservation of energy: carry a coulomb of charge once around the loop and back to where it started, and the energy it gained must equal the energy it gave up.

$$\sum_k V_k = 0$$



The sign rule. Pick a direction to travel the loop — clockwise or counter-clockwise, either is valid and both give the same answer. Then as you cross each component:

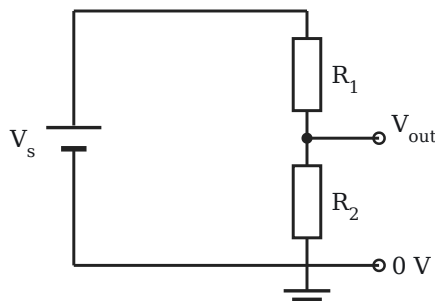
- entering at $-$ and leaving at $+$ is a **rise**: write $+V$;
- entering at $+$ and leaving at $-$ is a **drop**: write $-V$.

For the loop above this gives $V_s - V_1 - V_2 - V_3 = 0$, so $V_s = IR_1 + IR_2 + IR_3$, and therefore

$$R_{eq} = R_1 + R_2 + R_3$$

The same current flows through every element of a series string; it is the *voltage* that divides, in proportion to resistance. For two resistors that proportion is worth having at your fingertips:

$$V_{out} = V_s \cdot R_2 / (R_1 + R_2)$$



Read the figure with the formula in front of you: the same current $I = V_s / (R_1 + R_2)$ flows through both resistors, and $V_{out} = IR_2$. The fraction $R_2 / (R_1 + R_2)$ is just “this resistor's share of the total resistance”. Two quick checks on any answer: if the two resistors are equal you must get half of V_s , and V_{out} can never exceed V_s .

Careful: the divider formula assumes nothing else draws current from the middle node. Hang a load on that node and the ratio changes — which is exactly why a divider makes a poor power supply.

Problem 3

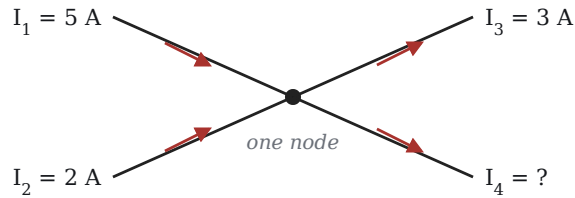
A 9 V source drives 1 k Ω , 2 k Ω and 3 k Ω in series.

- Find the current.
- Find the voltage across each resistor, and check that they sum to 9 V.
- Confirm the 2 k Ω drop using the divider formula, treating the other two as one resistance.
- Design a divider that turns 9 V into 3 V while drawing no more than 1 mA. Give both resistances.

4 KCL and the current divider

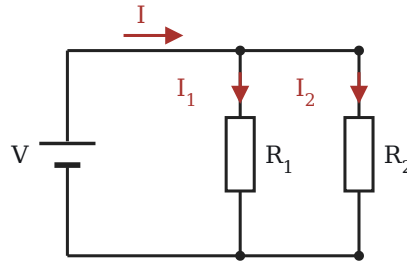
Kirchhoff's current law. Charge is conserved and does not pile up at a junction, so at every node the currents entering equal the currents leaving. Written with signs, the algebraic sum at a node is zero.

$$\sum_k I_k = 0 \quad \text{equivalently} \quad \sum I_{in} = \sum I_{out}$$



Apply it to two parallel resistors and you get the companion of the voltage divider:

$$I_1 = I \cdot R_2 / (R_1 + R_2)$$



the branch with the larger R takes the smaller share

Here both resistors see the same voltage V , so $I_1 = V/R_1$ and $I_2 = V/R_2$, and the two must add up to I . Eliminating V gives the formula above. Check it on the figure with $R_1 = R_2$: the current splits evenly, as it must.

Careful: the numerator is the *other* resistor. That is not a typo — current prefers the easier path, so the branch with the larger resistance gets the smaller share.

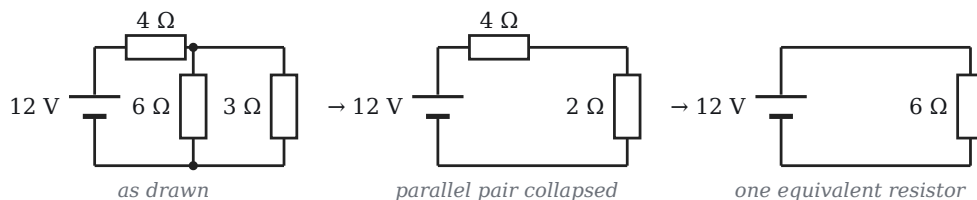
Problem 4

At a single node, 5 A and 2 A flow in, and 3 A flows out along a third wire.

- What is the current in the fourth wire, and is it entering or leaving?
- A student writes $5 + 2 + 3 + I_4 = 0$ and gets -10 A. What did the sign convention cost them, and what is the correct statement?

5 Putting it together — a worked example

Take a 12 V source driving a $4\ \Omega$ resistor in series with a parallel pair, $6\ \Omega$ and $3\ \Omega$. A network like this is handled by collapsing it from the far end back toward the source, then expanding outward again. Follow the three pictures left to right, then read the numbers underneath.



Collapse. The $6\ \Omega$ and $3\ \Omega$ share both of their nodes, so they are in parallel: $(6 \cdot 3)/(6+3) = 2\ \Omega$. That $2\ \Omega$ is in series with the $4\ \Omega$, so $R_{eq} = 6\ \Omega$ and the source delivers $I = 12/6 = 2$ A.

Expand. All 2 A flows through the $4\ \Omega$, so it drops $2 \cdot 4 = 8$ V. What is left, $12 - 8 = 4$ V, sits across the parallel pair — across *both* of them, since that is what parallel means. Ohm's law in each branch then gives $4/6 = 0.667$ A in the $6\ \Omega$

and $4/3 = 1.333\text{ A}$ in the 3Ω . Notice the smaller resistor took the larger current, and that the current divider gives the same numbers: $2 \cdot 3/(6+3) = 0.667\text{ A}$.

Power. $2^2 \cdot 4 = 16\text{ W}$ in the 4Ω , $4^2/6 = 2.67\text{ W}$ in the 6Ω , $4^2/3 = 5.33\text{ W}$ in the 3Ω .

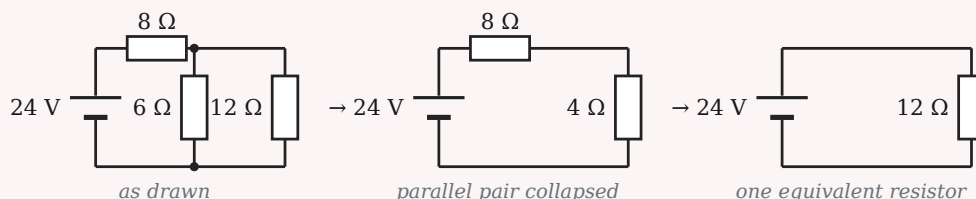
Then check three ways, every time:

- KCL at the top node of the pair: $0.667 + 1.333 = 2\text{ A}$, the total current. ✓
- KVL around the outer loop: $12 - 8 - 4 = 0$. ✓
- Power: $16 + 2.67 + 5.33 = 24\text{ W}$, and the source delivers $12\text{ V} \times 2\text{ A} = 24\text{ W}$. ✓

Two of the three are nearly free once you have the numbers, and together they catch almost every arithmetic error you are likely to make.

Problem 5

Same method, new numbers: a 24 V source drives an 8Ω resistor in series with a parallel pair of 6Ω and 12Ω .

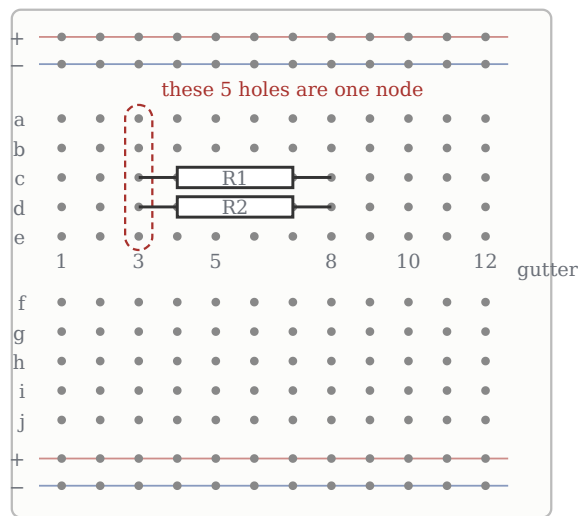


- Find R_{eq} and the total current.
- Find the voltage across the 8Ω resistor and across the parallel pair.
- Find the current in each branch, once with Ohm's law and once with the current divider.
- Find the power dissipated in each resistor.
- Run all three checks.
- The 8Ω resistor is replaced by 4Ω . Without redoing the whole problem, does the current in the 6Ω branch go up or down, and why?

6 The breadboard

A breadboard is a grid of spring clips wired together in fixed groups. Nothing about the plastic tells you which holes are connected, so you have to know the pattern:

- In the main area, each short strip of five holes is one node. In the orientation below, that is one *column* within a half.
- The long rails along the edges are one node each, running the whole length of the board. They are for power and ground — but they are connected to nothing until you run a wire to them.
- The gutter down the middle separates the two halves. Holes facing each other across it are *not* connected. That gap exists so a DIP chip can straddle it with each pin in its own node.



Two failure modes account for most of the debugging time in a first lab. Putting both legs of one component into the same five-hole group shorts that component out — the circuit behaves as if it were a plain wire. And assuming the rails are live when no wire runs from the supply to them leaves a circuit that is correct on paper and dead on the bench.

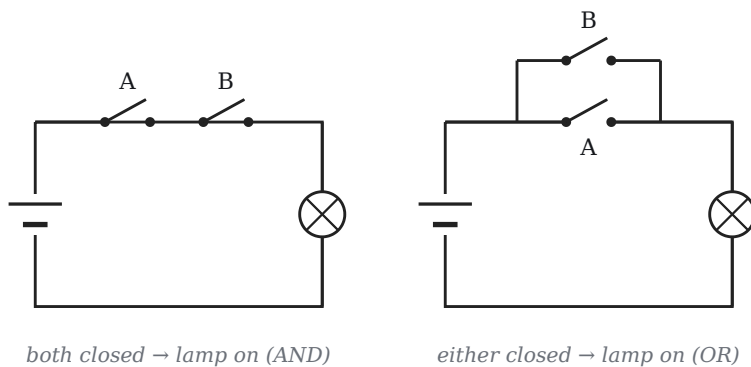
Problem 6

Read the figure above. Holes in the same column and the same half are one node.

- R1 spans columns 3 and 8 in row c; R2 spans the same two columns in row d. Are they in series or in parallel? Justify it with nodes, not with the picture.
- Is hole a8 connected to hole f8?
- A student inserts a third resistor with one leg in b6 and the other in e6. What have they built?
- Where would you connect the two ends of the pair so that a supply drives them?

7 From switches to logic

Two switches in series with a lamp: the lamp lights only if *both* are closed. Two in parallel: it lights if *either* is closed.



Now stop drawing lamps. Write 1 for “closed” and 0 for “open”, 1 for “lamp on” and 0 for “off”, and the two circuits become two tables — the truth tables of AND and OR. Everything from here on is algebra on those two symbols. The wires have not gone anywhere; we have simply stopped writing volts and amps, because for a digital circuit only the two states matter.

8 Boolean algebra: axioms and theorems

Boolean algebra is built the way Euclidean geometry is: a handful of axioms, everything else derived. Boole published it in 1849 as pure mathematics; it waited almost a century for switching circuits to make it the most useful algebra in engineering.

Axioms

- 1a $0 \cdot 0 = 0$
- 1b $1 + 1 = 1$
- 2a $1 \cdot 1 = 1$
- 2b $0 + 0 = 0$
- 3a $0 \cdot 1 = 1 \cdot 0 = 0$
- 3b $1 + 0 = 0 + 1 = 1$
- 4a if $x = 0$, then $\overline{x} = 1$
- 4b if $x = 1$, then $\overline{x} = 0$

Single-variable theorems

- 5a $x \cdot 0 = 0$
- 5b $x + 1 = 1$
- 6a $x \cdot 1 = x$
- 6b $x + 0 = x$
- 7a $x \cdot x = x$
- 7b $x + x = x$
- 8a $x \cdot \overline{x} = 0$
- 8b $x + \overline{x} = 1$
- 9 $\overline{\overline{x}} = x$

Two- and three-variable theorems

- | | | | |
|-----------------------------|--------------|--|-----------|
| 10a $x \cdot y = y \cdot x$ | Commutative | 14a $xy + \overline{xy} = x$ | Combining |
| 10b $x + y = y + x$ | | 14b $(x+y)(\overline{x+y}) = x$ | |
| 11a $x(yz) = (xy)z$ | Associative | 15a $\overline{x \cdot y} = \overline{x} + \overline{y}$ | DeMorgan |
| 11b $x + (y+z) = (x+y) + z$ | | 15b $\overline{x+y} = \overline{x} \cdot \overline{y}$ | |
| 12a $x(y+z) = xy + xz$ | Distributive | 16a $x + \overline{xy} = x + y$ | |
| 12b $x + yz = (x+y)(x+z)$ | | 16b $x(\overline{x+y}) = \overline{xy}$ | |
| 13a $x + xy = x$ | Absorption | 17a $xy + yz + \overline{xz} = xy + \overline{xz}$ | Consensus |
| 13b $x(x+y) = x$ | | 17b $(x+y)(y+z)(\overline{x+z}) = (x+y)(\overline{x+z})$ | |

Duality. Notice that everything comes in a/b pairs. Swap every $\cdot$ with $+$, every $+$ with $\cdot$, every 0 with 1 and every 1 with 0, and a true identity turns into another true identity — its dual. So the list is half as long as it looks: learn one of each pair and you can generate the other. (Duality is a statement about identities, not a licence to alter one side of an expression you are simplifying.)

Problem 7

Use the tables. Name the theorem you used at each step.

- Simplify $x(x+y)$.
- Simplify $x + \overline{xy}$.
- Write the dual of the distributive law $x(y+z) = xy + xz$, and find it in the table.
- Simplify $xy + \overline{xy} + \overline{xy}$.

9 Precedence of operations

In the absence of parentheses, the order is **NOT, then AND, then OR** — NAO. So $x + yz$ means $x + (yz)$, and $\overline{xy}$ means $(\overline{x})y$, not $\overline{xy}$. The bar is its own grouping symbol: everything under one bar is complemented together. When in doubt, write the parentheses; nobody has ever lost a point for an extra pair.

Problem 8

- Insert all the implied parentheses in $\overline{xy} + z$.
- Are $x + yz$ and $(x+y)z$ the same function? If not, give a set of values that distinguishes them.
- Are $\overline{x} \cdot \overline{y}$ and $\overline{x \cdot y}$ the same? Which theorem settles it?

10 Truth tables, and proving DeMorgan

A truth table settles any question about a Boolean identity: list every combination of the inputs, evaluate both sides, compare the columns. It is completely mechanical and always works. Its weakness is size — n variables need 2^n rows, so at six variables you are writing 64 lines. That is the motivation for the algebra: it gets the same answers without the exponential.

Here is 15a proved that way. The two right-hand columns match in every row, so the identity holds.

x	y	$x \cdot y$	$\overline{x \cdot y}$	$\overline{x}$	$\overline{y}$	$\overline{x+y}$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

In words: the complement of a product is the sum of the complements, and the complement of a sum is the product of the complements. Practically, DeMorgan is how you move inversions around a circuit — an AND gate with an inverted output is the same thing as an OR gate with inverted inputs, which is what makes the next section work.

Problem 9

- Prove 15b, $\overline{x+y} = \overline{x} \cdot \overline{y}$, with a truth table.
- How many rows would you need to check a three-variable identity? A six-variable one?

11 Algebraic proof and minimisation

Simplifying by algebra is a search, not a recipe, but four moves cover most of what you will meet:

- Multiply out first. Terms like $x\overline{x}$ then vanish by 8a, and pairs like $y+\overline{y}$ collapse to 1 by 8b.
- Factor with the distributive law to expose exactly those pairs.
- Use 7b, $x + x = x$, **backwards**: duplicate a term so it can pair with two different neighbours. Nothing is lost, since a term added to itself is itself. This is the step people never think of, and problem 12 cannot be done without it.
- To prove an identity, it is usually easiest to reduce *both* sides to the same expression, rather than forcing one to turn into the other.

Problem 10

Prove the identity:

- $(x_1+x_3)(\overline{x_1}+\overline{x_3}) = \overline{x_1}x_3 + x_1\overline{x_3}$
- What familiar two-input function is this?

Problem 11

Prove the identity by reducing each side separately:

$$\overline{x_1}x_3 + \overline{x_2}x_3 + x_1x_3 + \overline{x_2}x_3 = \overline{x_1}x_2 + x_1x_2 + \overline{x_1}x_2$$

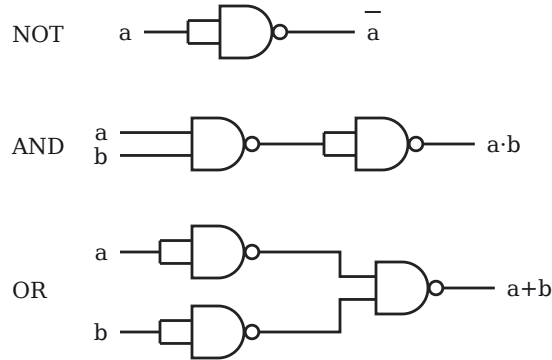
Problem 12

Minimise the function:

- $f(x_1, x_2) = x_1x_2 + \overline{x_1x_2} + \overline{x_1x_2}$
- Check your answer with a truth table.

12 NAND is a universal gate

Any Boolean function can be written as a sum of products, which needs only NOT, AND and OR. So if a single gate type can produce those three, it can produce *everything* — that is what “universal” means. NAND can, as can NOR:



NOT is a NAND with its inputs tied together, since $\overline{a \cdot a} = \overline{a}$. AND is a NAND followed by that inverter. OR is DeMorgan read backwards: invert both inputs, then NAND them, because $\overline{\overline{a} \cdot \overline{b}} = a + b$.

This matters commercially, not just theoretically: a fabrication process that is good at one gate type can build any logic at all, and in CMOS the NAND happens to be cheaper and faster than the AND.

Problem 13

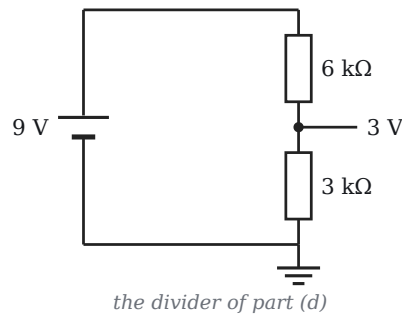
- Build XOR from NAND gates only. Four are enough. (Hint: start from $a\overline{b} + \overline{a}b$ and apply DeMorgan.)
- Show that NOR is also universal by building NOT, OR and AND from NOR gates.
- In the Vivado project from last time, what single character would you change in `assign y = a & b;` to turn your AND gate into an OR gate?

Solutions

Problem 1. (a) $R_{eq} = 12\Omega$, so $I = 12/12 = 1\text{ A}$; the drops are 4 V and 8 V. (b) Bottom = 0 V, middle = 8 V, top = 12 V. (c) Middle = 0 V, top = +4 V, bottom = -8 V. (d) Only the labels changed. The current is still 1 A and the voltage across each resistor is still 4 V and 8 V. Negative node potentials are perfectly ordinary; they only mean the node sits below the one you called zero.

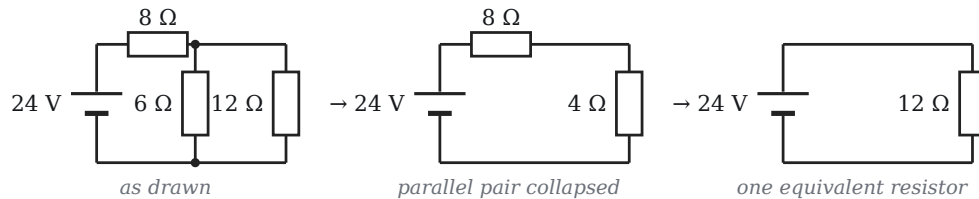
Problem 2. (a) $(6 \cdot 3)/(6+3) = 2\Omega$. (b) $3/3 = 1\Omega$. (c) $(100 \cdot 1)/101 = 0.990\Omega$ — essentially the 1Ω resistor alone; the 100Ω branch carries a hundredth of the current and hardly matters. (d) $1/4 = 1/12 + 1/R$ gives $1/R = 1/4 - 1/12 = 1/6$, so $R = 6\Omega$.

Problem 3. (a) $R_{eq} = 6\text{ k}\Omega$, $I = 9/6000 = 1.5\text{ mA}$. (b) 1.5 V, 3.0 V, 4.5 V; they sum to 9 V. (c) $9 \cdot 2\text{ k}/(2\text{ k}+4\text{ k}) = 3.0\text{ V}$, as found. (d) The ratio must be $R_2/(R_1+R_2) = 1/3$, i.e. $R_1 = 2R_2$. For $I \leq 1\text{ mA}$ the total must be at least $9\text{ k}\Omega$, so $R_1 = 6\text{ k}\Omega$ and $R_2 = 3\text{ k}\Omega$ is the limiting case; any larger pair in 2:1 ratio ($20\text{ k}\Omega$ and $10\text{ k}\Omega$, say) also works and wastes less current.

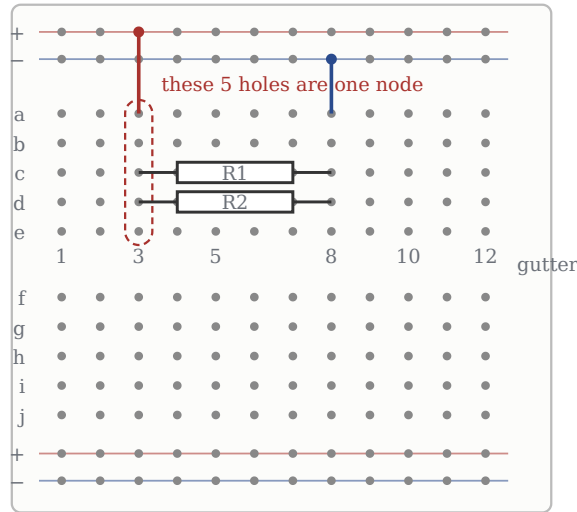


Problem 4. (a) $I_{in} = 5 + 2 = 7\text{ A}$, out so far = 3 A, so the fourth wire carries 4 A *leaving* the node. (b) They added four currents as though all four flowed the same way. The algebraic form requires a sign for direction: taking “in” as positive, $5 + 2 - 3 + I_4 = 0$, so $I_4 = -4\text{ A}$, i.e. 4 A leaving — the same answer. Either write $\sum I_{in} = \sum I_{out}$, or commit to one sign convention and keep it.

Problem 5. (a) $6||12 = (6 \cdot 12)/18 = 4\Omega$, $R_{eq} = 8 + 4 = 12\Omega$, $I = 24/12 = 2\text{ A}$. (b) $2 \cdot 8 = 16\text{ V}$ across the 8Ω ; $24 - 16 = 8\text{ V}$ across the pair (check: $2 \cdot 4 = 8\text{ V}$). (c) $8/6 = 1.333\text{ A}$ and $8/12 = 0.667\text{ A}$; by the divider, $2 \cdot 12/(6+12) = 1.333\text{ A}$ and $2 \cdot 6/(6+12) = 0.667\text{ A}$. (d) $2^2 \cdot 8 = 32\text{ W}$; $8^2/6 = 10.67\text{ W}$; $8^2/12 = 5.33\text{ W}$. (e) KCL: $1.333 + 0.667 = 2\text{ A}$. KVL: $24 - 16 - 8 = 0$. Power: $32 + 10.67 + 5.33 = 48\text{ W} = 24\text{ V} \times 2\text{ A}$. (f) Up. R_{eq} falls to 8Ω , so the total current rises to 3 A and the voltage across the pair rises to 12 V, giving 2 A in the 6Ω branch.



Problem 6. (a) Parallel. Both resistors have one leg in the column-3 node and the other in the column-8 node, so they share both nodes — the fact that they sit in different rows is irrelevant, because the five holes of a column are one piece of metal. (b) No. They are in opposite halves, separated by the gutter. (c) A short circuit across that resistor: b6 and e6 are the same node, so both legs are on the same wire and no current is forced through the part. (d) One wire from the + rail into column 3 and one from column 8 to the – rail — and remember to wire the rails themselves to the supply.



Problem 7. (a) $x(x+y) = x$ by 13b (absorption); or expand to $xx + xy = x + xy$ by 7a, then 13a. (b) $x + \overline{xy} = x + y$ by 16a. (c) The dual is $x + yz = (x+y)(x+z)$, which is 12b. Note this one has no counterpart in ordinary arithmetic — addition does not distribute over multiplication for numbers. (d) $xy + \overline{xy} = x$ by 14a, leaving $x + \overline{xy} = x + y$ by 16a.

Problem 8. (a) $((\overline{x}) \cdot y) + z$. (b) Not the same. Take $x = 1, y = 0, z = 0$: the first is $1 + 0 = 1$, the second is $(1+0) \cdot 0 = 0$. (c) Not the same: at $x = 1, y = 0$ the first is $0 \cdot 1 = 0$ and the second is $\overline{0} = 1$. DeMorgan (15a) gives the correct expansion, $\overline{xy} = \overline{x} + \overline{y}$.

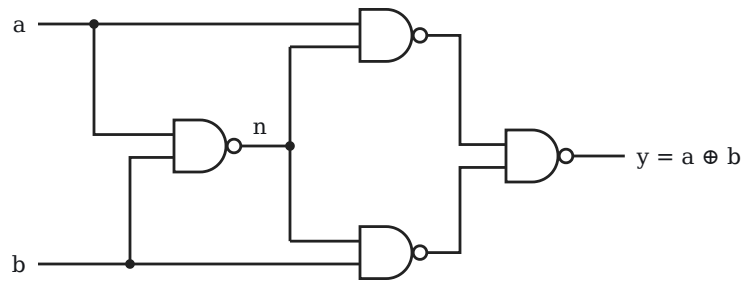
Problem 9. (a) The columns for $\overline{x+y}$ and $\overline{x} \cdot \overline{y}$ read 1, 0, 0, 0 down the four rows and therefore agree. (b) $2^3 = 8$ rows; $2^6 = 64$ rows.

Problem 10. (a) Expand with 12a: $(x_1+x_3)\overline{x_1} + (x_1+x_3)\overline{x_3} = x_1\overline{x_1} + x_3\overline{x_1} + x_1\overline{x_3} + x_3\overline{x_3}$. The first and last are 0 by 8a, and 6b removes them, leaving $x_1\overline{x_3} + \overline{x_1}x_3$. (b) XOR: true exactly when the two inputs differ.

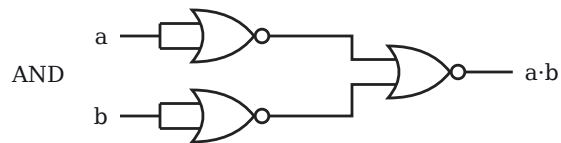
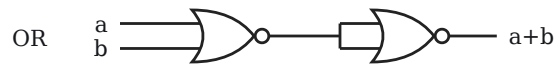
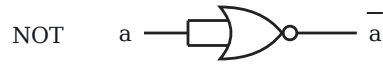
Problem 11. LHS: group the terms in pairs, $x_1(\overline{x_3}+x_3) + \overline{x_2}(\overline{x_3}+x_3)$ by 12a; each bracket is 1 by 8b; then 6a gives $x_1 + \overline{x_2}$. RHS: $\overline{x_1x_2} + x_1(x_2+\overline{x_2}) = \overline{x_1x_2} + x_1$ by 8b and 6a, which is $x_1 + \overline{x_1x_2}$ by 10b and then $x_1 + \overline{x_2}$ by 16a. Both sides equal $x_1 + \overline{x_2}$.

Problem 12. (a) Duplicate the last term using 7b, which is legal because a term plus itself is itself: $f = x_1x_2 + \overline{x_1}x_2 + \overline{x_1x_2} + \overline{x_1}x_2$. Factor the first pair on x_2 and the second pair on $\overline{x_1}$: $(x_1+\overline{x_1})x_2 + \overline{x_1}(\overline{x_2}+x_2) = x_2 + \overline{x_1}$ by 8b and 6a. (b) Both forms are 1 on the rows 00, 01, 11 and 0 on the row 10.

Problem 13. (a) With $n = \overline{ab}$ from the first gate, the output $\overline{(an) \cdot (bn)}$ is XOR; the four gates are the first NAND, two NANDs fed by (a,n) and (b,n) , and one NAND combining them. (b) NOT: tie the inputs of a NOR together. OR: a NOR followed by that inverter. AND: invert both inputs, then NOR them, since $\overline{a+b} = a \cdot b$. (c) Change $\&$ to $|$; the waveform for y becomes 0111.



(a) XOR from four NANDs



(b) NOT, OR and AND from NORs